

PIL (Python Imaging Library)

- => Imageelar bilan ishlaydigan asosiy librarylardan biri.
- => Image larini yuklaydi va ular ustida resize, crop kabi kiplar ishlatiladi. shulga oshirish imkonini beradi.

Tabular data: CSV file dan foydalanib pandas orqali ishlatiladigan (numpy) arraylar bilan ishlatiladi hamda Pytorch va Scikit-learn.

Image data: Image file - PIL/OpenCV - Numpy - Pytorch.

- => Numpy yoki tensorflow convert qilishdan oldin image preprocessing vazifasi uchun javobgar library.

Open CV library

Open Source Computer vision library.

Open Source => Posterning kodi hamma uchun ochiq: ishlatgan odam uni kiritishi, o'zgartirishi (o'zgarishlarni qo'shish) mumkin.

Computer vision (image va video) va image preprocessing uchun ishlatiladi.

- Autonomous vehicles
- Face recognition
- Medical imaging
- Security systems
- Robotics
- Augmented reality

kabi bōlimlarpla kōp iklōstīlināli.

Pillow (PIL)

lib type: Image processing

Main purpose: Loading and editing images

development: Python library (Pillow fork of PIL)

Speed: Moderate

Data structure: PIL Image Object

Color format: RGB

Image editing: Basic

Video processing: No

Computer vision Algorithms: No

Deep learning integration: Limited

Ease of use: Very easy

OpenCV

lib type: Computer vision library

Main purpose: Analyzing images and videos

development: C++ library with Python bindings

speed: Very fast

Data structure: Numpy array

color format: BGR

image editing: advanced

video processing: Yes

Deep learning integration: Strong

Ease of use: Slightly more complex